

Aidan Hennessey

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EDUCATION

2025 – 2030 Dartmouth College
PhD Math
2021 – 2025 Brown University
Sc.B. Math, A.B. Computer Science
2017 – 2021 Chelmsford High School

RESEARCH INTERESTS

Arithmetic geometry, combinatorial differential geometry

PAPERS

1. Many-Twist Möbius Bands With Small Aspect Ratios, *Comptes Rendus*, November 2024.

PREPRINTS

2. Computing Mod-5 Galois Images of Torsion on Abelian Surfaces, joint with Andy Zhu and Mathilde Kermorgant, [arXiv:2510.06423](https://arxiv.org/abs/2510.06423).
1. Tree and Tripod Nim, [arXiv:2401.07943](https://arxiv.org/abs/2401.07943).

TALKS

October	2025	Many-twist Möbius Bands, Dartmouth Undergraduate Math Society.
October	2025	Galois representations from elliptic curves, Dartmouth Graduate Student Seminar.
September	2025	Why can't you fold a piece of printer paper into a Möbius band? (Many-twist Möbius bands with small aspect ratios), Dartmouth Graduate Student Seminar.
May	2025	Many-twist Mobius bands with small aspect ratios, UPenn Graduate Student Geometry-Topology Seminar. Abstract .
March	2025	Using Elliptic Curves to Take Down Wolves in Sheep's Clothing, Brown University's Symposium for Undergraduates in the Mathematical Sciences (SUMS). Abstract (page 2).
October	2024	Galois Representations from Torsion of Abelian Surfaces, Brown Undergraduate Math Colloquium. Notes .
August	2024	Galois Representations from 5-Torsion of Abelian Surfaces, Young Mathematicians' Conference. Abstract (page 11).
March	2024	Many-Twist Möbius Bands, Brown University's Symposium for Undergraduates in the Mathematical Sciences (SUMS). Abstract .
February	2024	Nim Arrays and Tripod Nim, Brown Undergraduate Math Colloquium. Slides .

SELECTED GRANTS AND AWARDS

May	2025	Howell Premium Prize, \$1000
April	2024	Henry Parker Manning Prize, \$300
June	2023	Brown Undergraduate Teaching and Research Award (UTRA), \$2500

MATH INSTRUCTION

DARTMOUTH COLLEGE

Winter	2026	TA, Math 22: Linear Algebra with Applications
Fall	2025	TA, Math 1: Introduction to Calculus

BROWN UNIVERSITY

Spring	2025	TA, Math 1710: Topology
Spring	2024	TA, Math 1580: Cryptography
Fall	2023	TA, Math 1530: Abstract Algebra

VARSIITY TUTORS

2023 – 2024	Competition Math Tutor (AMC 10/12)
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CHELMSFORD HIGH SCHOOL

2017 – 2021	Peer Tutor
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MATHNASIUM OF ACTON

2020	AMC 8 Workshop Instructor
2020	Tutor (Prealgebra through Calculus)

NOTABLE MATHEMATICS COURSEWORK

Learning seminars on abelian varieties, the Weil conjectures, and Teichmüller theory, 4 semesters of algebra (2 undergrad, 2 grad), 2 semesters of number theory (1 undergrad, 1 grad), 4 semesters of topology (1 undergrad, 3 grad, 1 topics), 2 semesters of manifolds (1 undergrad, 1 grad), graduate algebraic geometry, graduate complex analysis, real analysis, applied PDEs, differential geometry, information theory, graph theory.

NOTABLE PROGRAMMING EXPERIENCE

Summer	2024	Summer research in computational arithmetic geometry. The end result of this project was an over 1000 line program written in the C-based computer algebra language Magma. Publicly available on github .
Summer	2023	Summer research in combinatorics, heavily reliant on numerical experiments in Python.
Winter	2023	Built an Ascii implementation of the game Amazons, supporting both two-player gameplay and solo play against a strong artificial intelligence using Monte-Carlo tree search. Available for download on github .
Fall	2022	Systems Programming course in C, including networks and parallel computing.
Spring	2022	Data Structures and Algorithms course in Java
Spring	2022	Computer Vision course featuring heavy use of Python, specifically libraries NumPy and Tensorflow. Culminated in final project, viewable here .